

# LÊ GIA HÙNG

0966148654 | [legiahung2908@gmail.com](mailto:legiahung2908@gmail.com) | <https://github.com/LeGiaHung29805>

## EDUCATION

---

HaNoi University of Natural Resources and Environment

Sep 2023 – Present

Major: Information Technology | GPA: 3.99

## SOFTWARE DEVELOPMENT EXPERIENCE

---

Academic Software Development Team, HUNRE

Jul 2025 – Present

Software Engineer Intern

- **Full-stack Web:** Engineered an e-commerce platform using React/Next.js and Java (Spring Boot). Independently managed server deployment, ensuring high availability and seamless transactions.
- **Desktop Automation:** Developed C# (.NET WinForms) utility tools to automate complex gaming operations. Optimized execution logic and GUIs, significantly reducing manual interaction time.

## RESEARCH & DEVELOPMENT PROJECTS

---

Analysis and Optimization of Evacuation Routes  
and Shelters in Flood Conditions in Lao Cai

Jan 2026 – Apr 2026

- **Architected a Hybrid Deep Learning pipeline** (Stacked LSTM & Random Forest) with an Asymmetric Trade-off prioritizing Recall (81%) for human safety. Implemented Tensor Reshaping, SMOTE, and Cyclical Feature Encoding to resolve class imbalance and temporal blindness, achieving 94% Accuracy and 0.9697 ROC-AUC.
- **Engineered a high-performance Spatial Data Fusion engine** integrating Mask Sampling to transform 2.5D rasters into 1D feature vectors. Optimized spatial queries via Euclidean Distance Transform (EDT), reducing algorithmic complexity from  $O(N \times M)$  to  $O(N)$  and slashing processing time from hours to seconds.
- **Developed a Risk-aware Dijkstra routing algorithm** over a Topo-Multigraph with Dynamic Cost assignment. Integrated AHP and Spatial Carrying Capacity theory to calculate optimal evacuation routes under Worst-case Scenario Inferences, achieving soft real-time convergence (20-30ms latency).
- **Led end-to-end system deployment**, Established an Explainable AI (XAI) framework by transforming expert planning into a Rule-based System and decoding Gini Impurity to provide transparent, auditable disaster response recommendations.

## TECHNICAL SKILLS

---

**Programming Languages:** Python, Java, JavaScript, TypeScript.

**AI/DL & ML:** PyTorch, TensorFlow, Scikit-learn, Imbalanced-learn, OpenCV.

**GIS & GeoAI:** GeoPandas, Rasterio, OSMnx, Shapely, Google Earth Engine.

**Web, Frameworks & Architecture:** FastAPI, Spring Boot, React/Next.js, Microservices.

**Databases & DevOps (MLOps):** PostgreSQL/PostGIS, Redis, Docker (Compose), Git, SQLAlchemy.